
Type of fake news prediction in French

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Abstract

We analyse a dataset containing different label linked to types of fake news in French. We begin by analysing the dataset which, labelled by several people, contains a lot of contradictory information. We then analyse the various state-of-the-art methods for multilabel analysis. As these methods are difficult to use with our dataset, we then propose a few methods to analyse the different labels. We finally try to use a state of the art llm to see if it can be useful for our problem.

1 Data analysis

Our dataset is made up of 100 lines containing information on 100 texts from information or pseudo information websites. For each text we have the link and 11 labels filled in. Each text is labelled by 8 different people. We'll see in the next section that this is useful for getting a better idea of the different criteria, but raises a number of questions about the quality of the labelling.

1.1 Data scraping

As the public dataset did not contain the texts, the first step was to scrape the websites to retrieve the texts. This first step was a source of disappointment, as the texts from the 'moutonsrebelles' site are no longer available online. This leaves us with just 93 texts.

We are also making a first approximation here since we consider that all the information is contained in the texts. However, the labels relate to the entire articles (text + photos). As the sources and specific data can sometimes be found on pictures, we will not be able to find them in the texts and therefore in our analysis.

We provide the code used for the scrapping (two methods) in notebook scrapping, but we would like to point out that we have compared the two methods in order to retain the best result of the two.

1.2 Data analysis

As indicated at the beginning of this section, each text is labelled by 8 different people. We will see in this section that they generally disagree with each other. Here, we use a different disagreement measure than the one used in the original paper.

To analyse the agreement between the different people who have labelled the dataset, we construct a disagreement indicator. For each text and for each label, we note p the average of the labels indicated by the 8 people. We then use the following indicator $I = 4p(1 - p)$. So if all the people agree this indicator will be worth 0 and if 4 people have indicated 1 and 4 people have indicated 0 (maximum disagreement) the indicator will be worth 1.

We thus present the bar chart by category containing the disagreement score presentation for every category and every text. Here we see the second biggest problem of the dataset after its relatively small size: raters are generally in little agreement with each other on many categories. This is particularly marked in certain cases, such as the "insinuation", "sources cited" and "reported information" categories. This makes analysis particularly difficult for our future work, as we are not sure whether the text actually contains a feature or not. It is also interesting to note that for the "fake news" category, which is supposed to be the main category in the dataset, there are only around thirty texts where the raters all agree with each other. This underlines the particularity of this field: it is sometimes very

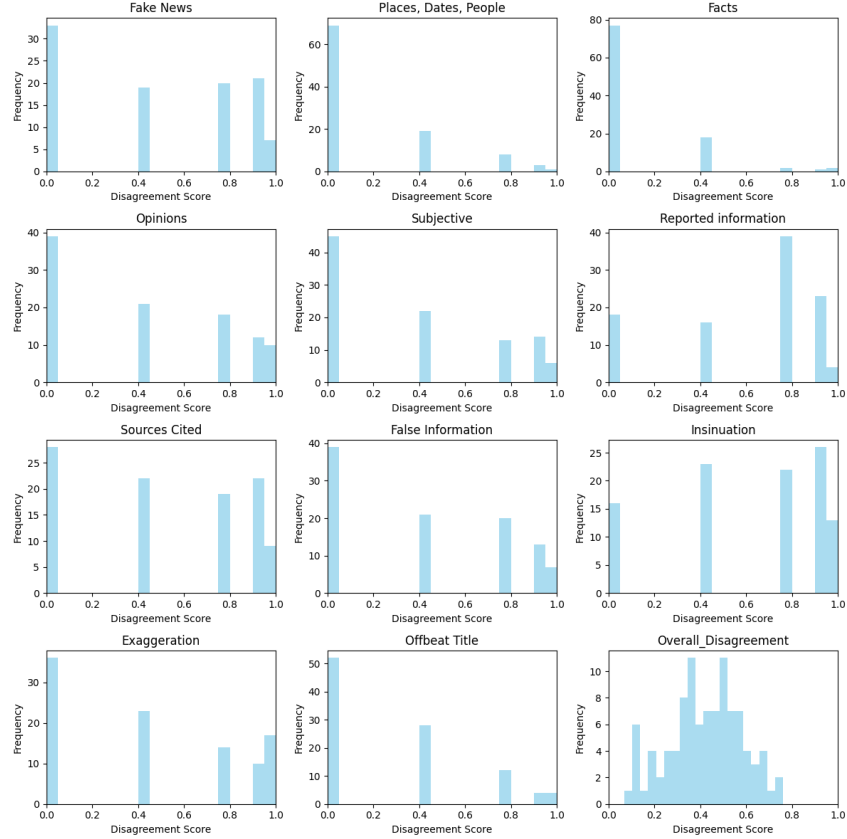


Figure 1: Disagreement score by label

difficult, even for a human in a privileged setting, to detect fake news.

We therefore had to make a choice for our study. We will take the majority opinion among the raters as the analysis label for the rest of the report. In the case of a 4-4 equality, we will act as if the text contained this characteristic, thus keeping the value 1. This is a choice on our part which is motivated by the fact that we consider that it is more likely not to detect a characteristic than to detect one erroneously. This working hypothesis is questionable but seems justified for our study, we could also have chosen randomly between 1 and 0 for the cases of equality. We present the overall situation for each label

We will not consider the “Facts” label and the “Places, Dates, people” label below, as they are

	Fake News	Places, Dates, People	Facts	Opinions	Subjective	Reported information	Sources Cited	False Information	Insinuation	Exaggeration	Offbeat Title
0s	56	0	0	17	29	79	17	73	37	39	88
1s	37	93	93	76	64	14	76	20	56	54	5

Figure 2: Label distribution

included in our entire dataset. There are also some surprising points in this presentation of labels: 37 texts are considered to be fake news, while only 20 texts contain false information. We found it difficult to justify that fake news does not contain any false information. We will therefore consider 9 labels for the rest of our study.

2 State of the art

2.1 Multi label analysis

Multi-labelling has been a subject of discussion for more than a decade. Even before the 2010s, [TK09] mentions the various challenges and the different methods put in place at the time to perform

multi-label analysis. This article lays the foundations: metrics to consider, possible challenges. Although the points of interest presented in the paper are not up to date, it remains important because the various challenges are still the same.

Techniques have evolved considerably in recent times. Some authors like [JLC20] use TF-IDF methods while others like [Bro+14] use simpler methods like bag of words. These methods are effective in many contexts but can be limited in situations where the labels are more subtle. In our case, think of the labels insinuation or exaggeration, both of which require contextual awareness that these two methods cannot provide.

Trying to do multi-label classification in different languages is a particularly popular topic. The use cases are particularly numerous. For example, [Al-24] classifies medical documents in Arabic to increase productivity in hospitals, while [Nab+24] uses similar methods in Indonesian for sentiment analysis. One strategy in particular seems to stand out from the others: use of a pre-trained model such as Bert and fine tuning with various methods. We will also apply this method in our empirical method.

2.2 Fake news analysis

The literature on detecting fake news is particularly well developed. [Roh+22] highlights the diversity of possible and developed methods for detecting fake news. There are also a number of papers that focus specifically on the problem that we are going to address here. For example, [HA21] studies the different types of fake news in Arabic linked to the covid 19 crisis.

However, it is important to point out that all the articles we have highlighted have much larger databases. Our study will be particularly limited by this fact. This is not simplified by the fact that our texts are considerably longer than in the various studies we have cited above. Although this larger text size makes it possible to extract more subtle characteristics, it also makes training considerably more complicated.

3 Empirical strategies & results

In the first part we will use Bert, and more specifically the model pre-trained on French data camemBERT. We will then use an MLP that we will train on our database. This fairly classic method, which we have already mentioned in the literature review, generally works quite well with longer corpus, so this will be an opportunity to work on a smaller dataset. In the third part, we will use a large language model to see whether this tool, which is clearly over-dimensioned for our task, can prove effective. This may prove particularly useful in our case, as we have very little data. In both cases, the models are able to understand the context, which should allow them, at least in theory, to capture more subtle aspects such as exaggeration or insinuation.

3.1 Bert

We Describe the strategy we use in this part in figure 3

3.1.1 Data Cleaning

This part is generally rather important in the literature. There are generally many characters that the model will not be able to understand (emoji, slang or abbreviated language). We're lucky in this respect, as our dataset is fairly clean. Because of the origin of the text (even the conspiracy sites want to be considered as serious thus tend to use very little slang) there are very few characters like emojis and very little abbreviated language. The only source of cleaning that we have to do in the dataset comes from the recovery of the texts that we have done. There may be unwanted html tags or spaces embedded. We will deal with this before beginning our study.

3.1.2 Bert and MLP parameters

There are a few points of interest worth considering before describing in more detail the method we're going to use in this section:

First of all, let's start with the point we made earlier about labels not being particularly balanced in our dataset. We're not in case of the [BS19] article and therefore have no particular interest in data

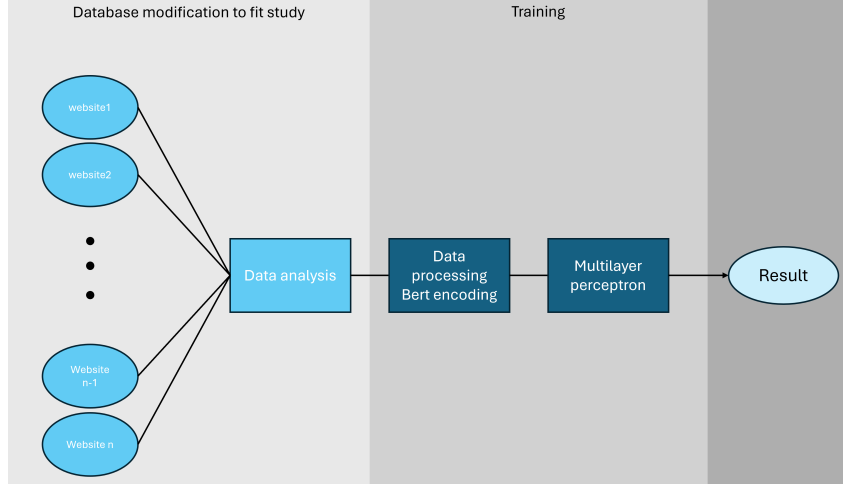


Figure 3: Empirical Strategy

augmentation linked to label. Instead, we choose to modify the weights in the loss function for each label. We do this by counting the number of positive values for each label and giving greater weight to labels that have few or even very few positive labels.

We combine these weights with a binary cross entropy loss function with logits loss. As our labels are mainly independents between one another this function is adapted. We use the version already in PyTorch with both the sigmoid function and binary cross entropy because it is more numerically stable than doing one and then the other.

For the Multilayer perceptron, we use 4 layers :

1. **Linear Transformation:** This layer projects the high-dimensional BERT embedding $x \in \mathbb{R}^{d_{\text{BERT}}}$ into a lower-dimensional hidden space of size d_{mlp} .
2. **ReLU Activation:** A ReLU non-linear activation function is applied element-wise to introduce non-linearity into the model.
3. **Dropout Regularization:** A dropout layer is used to randomly zero out a fraction of the activations during training to prevent overfitting.
4. **Final Linear Transformation:** The dropped-out activations are mapped to the output space corresponding to the number of labels n .

This method gives us a relatively disappointing result: the algorithm has a very poor ability to generalise, which is undoubtedly due to the lack of text in the dataset. Exact accuracy is not really the best metric to look at here, so in this section we are more interested in the f1 score and the hamming score. We find a hamming score of around 0.50, which is not that much better than the random classifier. We show the result in figure 5

In order to solve the problem we have here, we propose to implement a “data augmentation” technique. To do this, we will use the following pipeline: for each text, we will use gpt 4.1 mini to translate the text into English. We then use gpt 4.1 to translate the text back into French. We’re making the working assumption here that the style of the writing won’t be altered too much by this passage. This could prove complicated for certain categories such as ‘exaggeration’ or ‘insinuation’.

This technique has been quite popular for some times but is proving to not be the most effective technique possible with large language model. [Zha+24] explore a rewriting method (close to the one we are using right now) and a method where they create new data from the previous texts. The latest method seemed more effective but is quite hard to see here as the label of the newly created text will be very hard to predict. One thing that would have been possible to do is to make a llm to create new text with precise characteristics. "Create a fake news text where some conspiracy theory are implied but not directly written,...".

3.2 Data augmentation

After doing the data augmentation, we effectively doubled the number of rows we have to consider. However, here we have to be careful not to overfit the model. In the case of data augmentation by double translation, we always have to consider the created data and the real data differently. In our case we need to be particularly careful. This is due to the size of our texts, which are so large that despite a double translation, the original version and the double-translated version have a lot in common. Training an algorithm on data knowing that their translated data is in the validation or test database would mean not being able to really establish the model's performance. We therefore separate the data by 'URL', and a 'URL' (and therefore an article) can only be in one of the following categories at a time: train, validation and test. This prevents us from having an algorithm that overfits because of data leakage, but still allows us to learn better because the algorithm will have seen more different turns of phrase.

We want to assess the quality of the texts generated, as there would be no point in training our model on twice as much data but with absolutely identical characteristics. To evaluate the differences between the models we're using BERT again, we're going to use tokenisation and then run a two-component pca on the tokens to visualise the differences between the texts. We found in figure 4 that the translated texts are close but not exactly the same that their original version which is exactly what we need for our study.

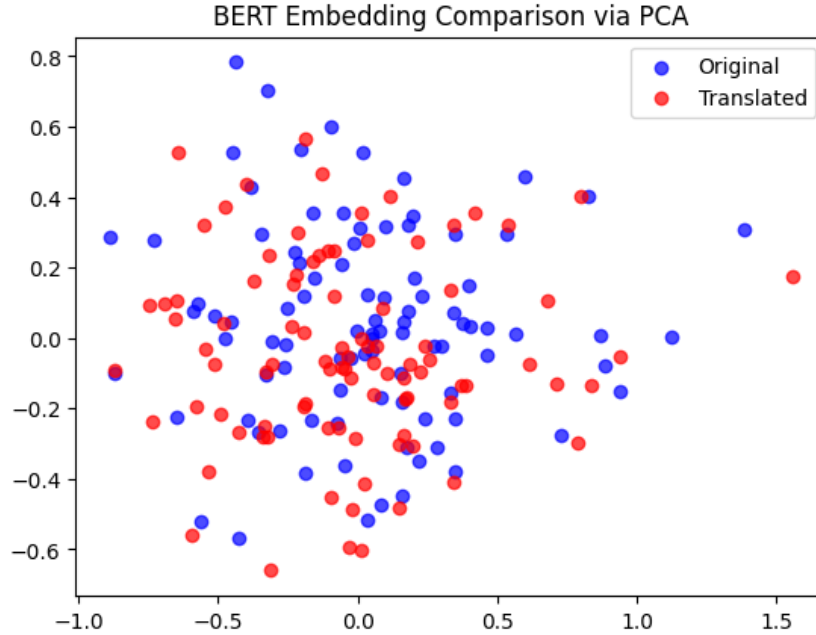


Figure 4: PCA Analysis between translated text and original text

We present our result in figure 5.

	Hamming Score	F1 score - micro	F1 score - macro
MLP without data augmentation	0.46	0.50	0.45
MLP with data augmentation	0.28	0.67	0.62

Figure 5: Metrics of MLP method both before and after data augmentation

3.3 What about LLM ?

In the previous section we had problems with the generalisation following our training. This was mainly due to the lack of data in our model, and one way of solving this would be to use a model that has already been trained on a lot of data and therefore doesn't need to determine which elements are likely to be signs of exaggeration or innuendo. We therefore propose to re-use the pipeline set up for the data augmentation part in order to test whether it would be possible, using various techniques, to take direct advantage of the llm's capabilities.

Here, we adopted a systematic approach to prompt engineering to optimize the performance of the LLM we used. We began by articulating a clear definition of the task and the desired output, ensuring that the prompt provided unambiguous and specific instructions to the model. To achieve this, we initially composed prompts that precisely detailed the expected format and reasoning steps, with the aim of encouraging the model to produce thoughtful and contextually appropriate responses. Even with that done, it sometimes gave result out of the format we wanted. We then iteratively refined these prompts by analyzing the generated outputs, identifying areas where ambiguity or insufficient guidance led to suboptimal performance (some labels were really badly predicted because the question was quite not clear for the llm, leading it to answer only 1 or only 0). During this process, we experimented with the inclusion of representative examples and additional context to direct the model's responses further and to reduce noise. The iterative adjustments allowed us to fine-tune the balance between clarity and comprehensiveness, resulting in prompts that gave detailed response.

The results we are seeing are very encouraging. After fine tuning the prompt, we found a hamming score close to 0.25, which is much better than the strategy we used previously without data augmentation. I doubt we could really obtain a better hamming score for the reasons we mentioned in the data analysis section. Indeed, as we saw in the previous section, many of the labels are ambiguous and we had to restrict ourselves to taking the majority opinion despite the very different opinions of the misses. We could improve the model's performance by having it tested several times in several languages. The subtleties of the different languages regarding exaggeration or insinuation would enable us to obtain a more objective result on average (provided that these two characteristics are well preserved during translation). We would, however, go beyond the scope of the experiment we had set ourselves for this report, the objective being to work on a French database.

4 Comments and conclusion

In this short article, we took advantage of the dataset provided in the "Type of fake news prediction in French" project proposal to explore a classic but effective method in multi-label classification. We faced the inherent limitation of a dataset containing very little data and whose different labels were not extremely clear. These types of labels, whose interpretation is even difficult for human, are particularly interesting for this kind of subject since it is a field where artificial intelligence and especially NLP can be most effective. We also had to overcome the fact that the dataset had too little data to be trained on, so we had to implement data augmentation techniques and evaluate, at least briefly, their effectiveness to verify that they were useful. Once the dataset was augmented, we obtained significantly more satisfactory results.

We then compared the effectiveness of our model with a state-of-the-art LLM model, namely GPT 4.1 mini. This model, not needing to learn what exaggeration or insinuation is, proves directly very effective. However, we can see that its effectiveness is not overwhelming compared to our model with augmented data, highlighting that the technique implemented can be very promising with a larger dataset.

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